Multi-species Catalogues of Regulatory Elements: Ensembl

Session 3

T6: Making the most of cis-regulatory annotations for genome-scale analysis

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EMBL-EBI

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03/09/2026

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Session map

The Ensembl Project	• General description
Ensembl Regulation	• Regulatory pipelines • Available datasets and Regulatory annotation • Defining and accessing reference epigenome data
The Ensembl Regulation subsite	• Epigenomes data and metadata • The regulatory features tab • Exploring genome annotation and regulatory activity
Comparison of CREs frameworks	• Most-popular genome-wide resources • Strengths and weaknesses of each approach
Hands-on	

The Ensembl Project

The mission of Ensembl is to provide comprehensive and accurate annotation of genomes, integrating this information with other biological data and making it publicly accessible via the web.

ENSEMBL

Have you used Ensembl before?

Using genome browsers to visualise annotation

CGGCCTTTGGGCTCGGCCTTCAGCTCAAGACTTAACTTCCCTCCCAGCTGTCCCAGATGACGCCATCTGAAATTTCTTGGAAACACGATCAC

Regulatory element

TTTAACGGAATATTGCTGTTTTGGGGAAAGTGTTTTACAGCTGCTGGGCACGCTGTATTTGCCTTACTTAAGCCCCTGGTAATTGCTGTATTC

CGA**AGACATGCTGATGGGAATTA**CCAGGCGGCGTTGGTCTCTAACTGGAGCCCTCTGTCCCCACTAGCCACGCGTCACTGGTTAGCGTGATT

GAAACTAAATCGTATGAAAATCCTCTTCTCTAGTCGCACTAGCCACGTTTCGAGTGCTTAATGTGGCTAGTGGCACCGGTTTGGACAGCACA

GCTGTAAAATGTTCCCATCCTCACAGTAAGCTGTTACCGTTCCAGGAGATGGGACTGAATTAGAATTCAAACAAATTTCCAGCGCTTCTGA

Gene

GTTTTACCTCAGTCACATAATAAGGAATGCATCCCTGTGTAAGTGCATTTTGGTCTTCTGTTTTGCAGACTTATTTACCAAGCATTGGAGGA

ATATCGTAGGTAAAA**ATGCCTATTGGATCCAAAGAGAGGCCAACATTTTTTGAAATTTTTAAGACACGCTGCAACAAAGCAGGTATTGACA**

AATTTTATATAACTTTATAAATTACACCGAGAAAGTGTTTTCTAAAAAATGCTTGCTAAAAACCCAGTACGTCACAGTGTTGCTTAGAAC

CATAAACTGTTCCTTATGTGTGTATAAATCCAGTTAACAACATAATCATCGTTTGCAGGTTAACCACATGATAAATATAGAACGTCTAGT

GGATAAAGAGGAAACTGGCCCCTTGACTAGCAGTAGGAACAATTACTAACAAATCAGAAGCATTAATGTTACTTTATGGCAGAAGTTGTC

CAACTTTTTGGTTTCAGTACTCCTTATACTCTTAAAAATGATCTAGGACCCCCGGAGTGCTTTTGTTTATGTAGCTTACCATATTAGAAA

TTTAAAACTAAGAATTTAAGGCTGGGCGTGGTGGCTCACGCCTGTAATCCCAGCACTTTGGGAGGCCGAGGTGGGCGGATCACTTGAGGC

CAGAAGTTTGAGACCAGCCTGGCCAACATGGTGAAACCCTATCTCTACTAAAAATACAAAAAATGTGCTGCGTGTGGTGGTGCGTGCCTGT

AATCCCAGCTACACGGGAGGTGGAGGCAGGAGAATCGCTTGAACCCTGGAGGCAGAGGTTGCAGTGAGCCAAGATCATGCCACTGCACTCTA

GCCTGGGCCACATAGCATGACTCTGTCTCAAAACAAACAAACAAACAAAAAACTAAGAATTTAAAGTTAATTTACTTAAAAATAATGAAAGC

TAACCCATTGCATATTATCACAACATTCTTAGGAAAAATAACTTTTTGAAAACAAGTGAGTGGAATAGTTTTACATTTTTGCAGTTCTCTT

TAATGTCTGGCTAAATAGAGATAGCTGGATTCAACTTATCTGTGTCTAATCTGTTATTTTGGTAGAAGTATGTGAAAAAAAATTAACCTCACC

Variant

TTGAAAAAAGGAATATTTTAATAGTTTTCAGTTTACTTTTTGGTATTTTTCCTTGTACTTTGCATAGATTTTTCAAAGATCTAATAGATATAC

CATAGGTCTTTCCCATGTCGCAACATTCATGCAGTGATTATTTGGAAGATAGTGGTGTTCTGAATTA

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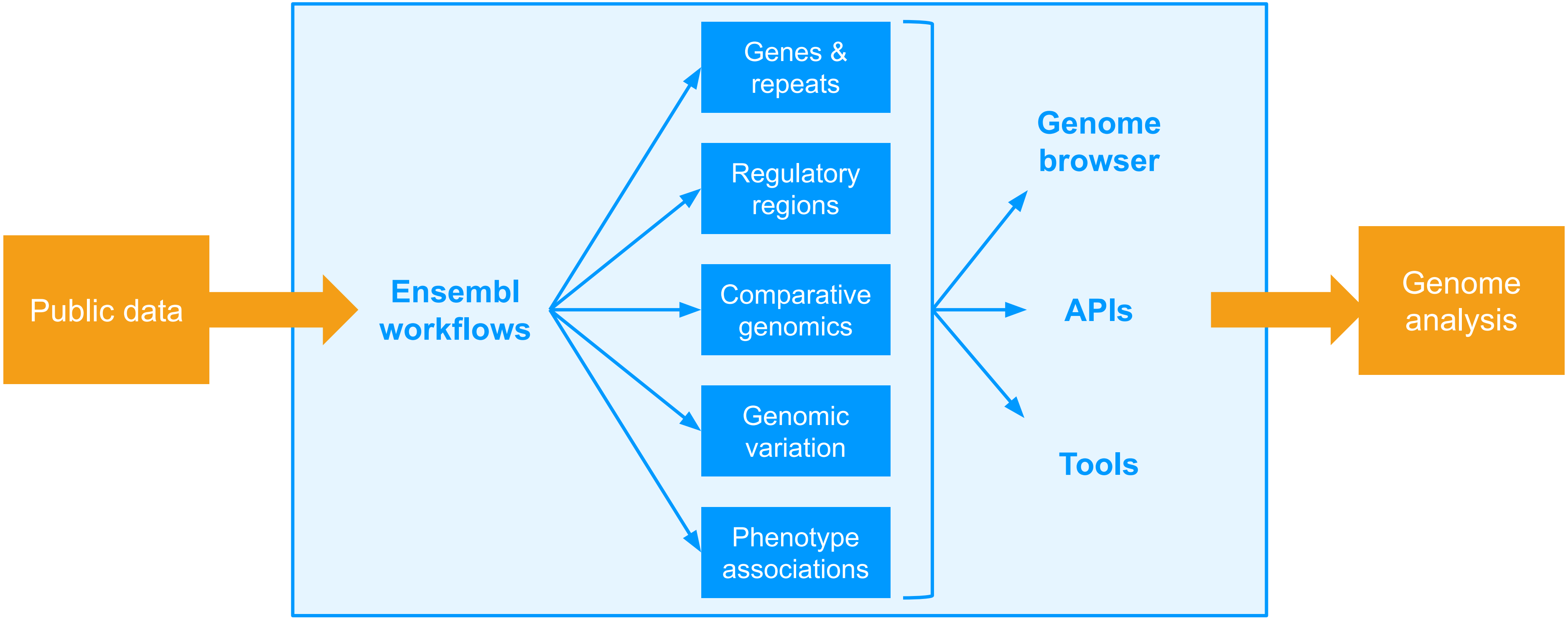
Ensembl enables genome analyses



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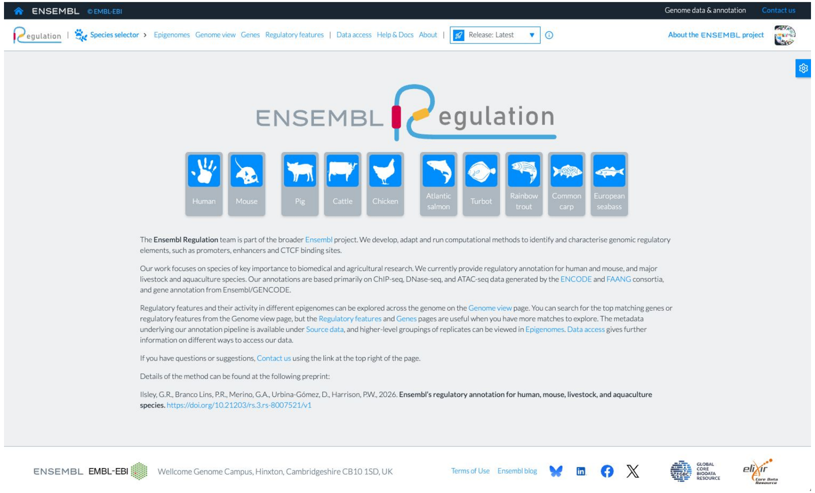
Ensembl Regulation

- ✓ Aim: Develop, adapt and run computational methods to identify and characterise genomic regulatory elements, such as promoters, enhancers and CTCF binding sites.
- ✓ Focus: Species of key importance to biomedical and agricultural research.
- ✓ Annotations are based primarily on ChIP-seq, DNase-seq, and ATAC-seq data generated by the ENCODE and FAANG consortia, and gene annotation from Ensembl/GENCODE.



https://regulation.ensembl.org/

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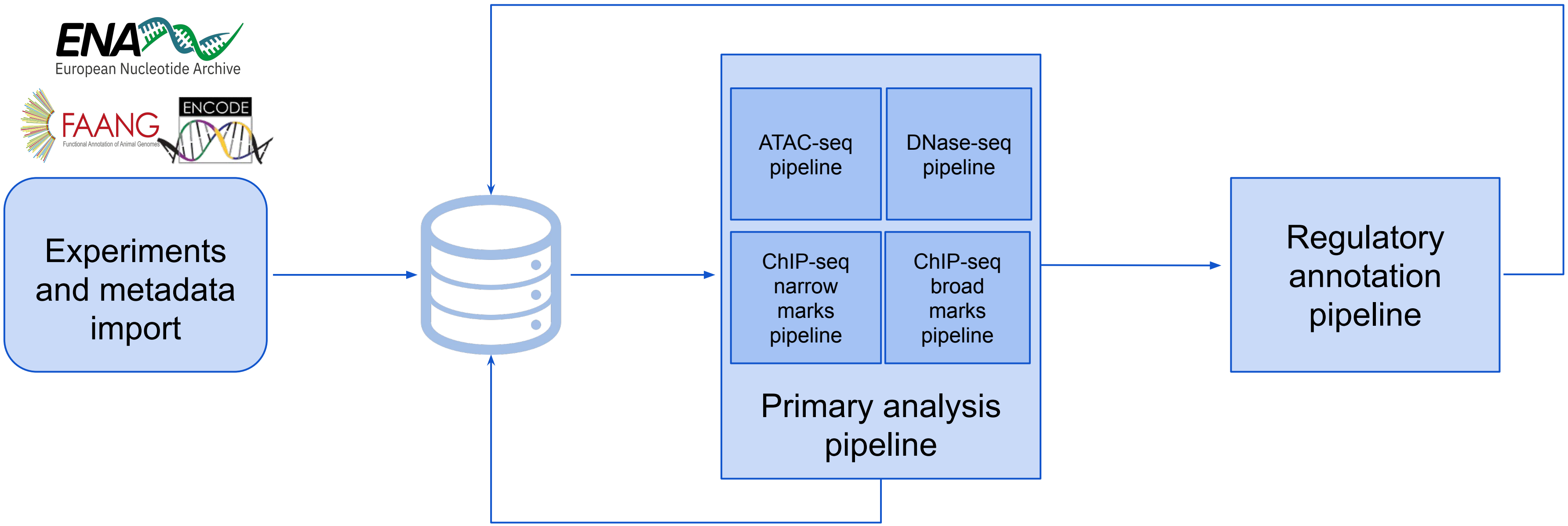
The Ensembl Regulatory Build



Primary analysis pipeline

✓ Processes read files to produce peak (BED) and signal (bigWig) files.

Preprocessing → Alignment → Peak calling → Signal files

- Preprocessing: Adapters trimming with NGmerge/fastp.
- Alignment: Bowtie2.
- Peak calling:
    - Genome mask:
        - ENCODE masks for human and mouse.
        - Custom masking strategy for other species.
    - Genrich with different configurations for each experiment type.
- Signal: Estimated from Genrich pileups

Standardised data analysis pipelines

https://regulation.ensembl.org/help/primary_processing

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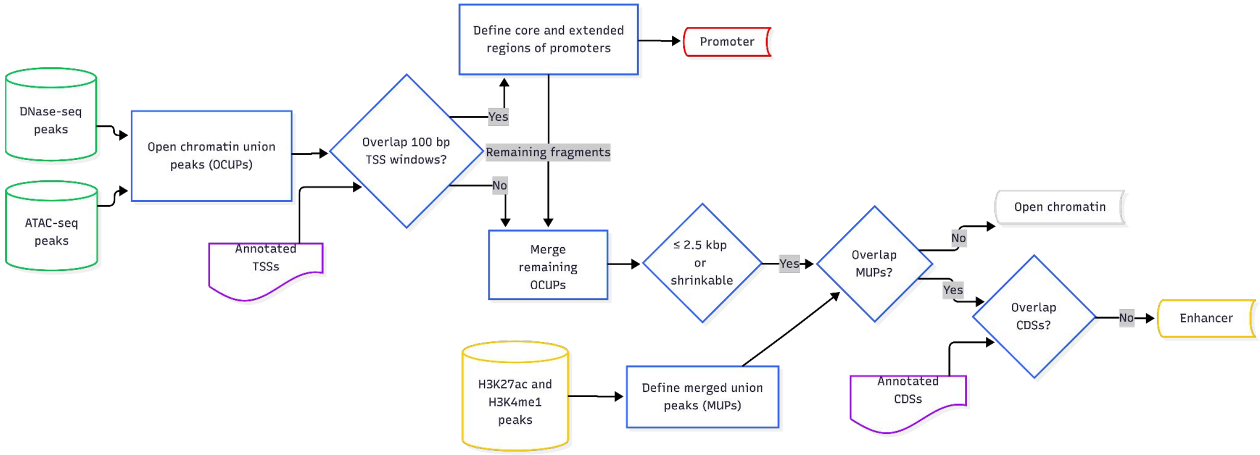
Regulatory annotation pipeline



Input:

- **ATAC-seq & DNase-seq** peaks for all the epigenomes of the target species
- H3K27ac, H3K4me1, and H3K4me3 ChIP-seq peaks for all the epigenomes of the target species (**optional**)
- Annotated TSSs and CDSs from transcripts of interest

Output:

- Set of regulatory features: **Promoters, Enhancers, OCR**
- Regulatory features (RFs) **activity** across different epigenomes with open **chromatin data**

https://regulation.ensembl.org/help/regulatory_build

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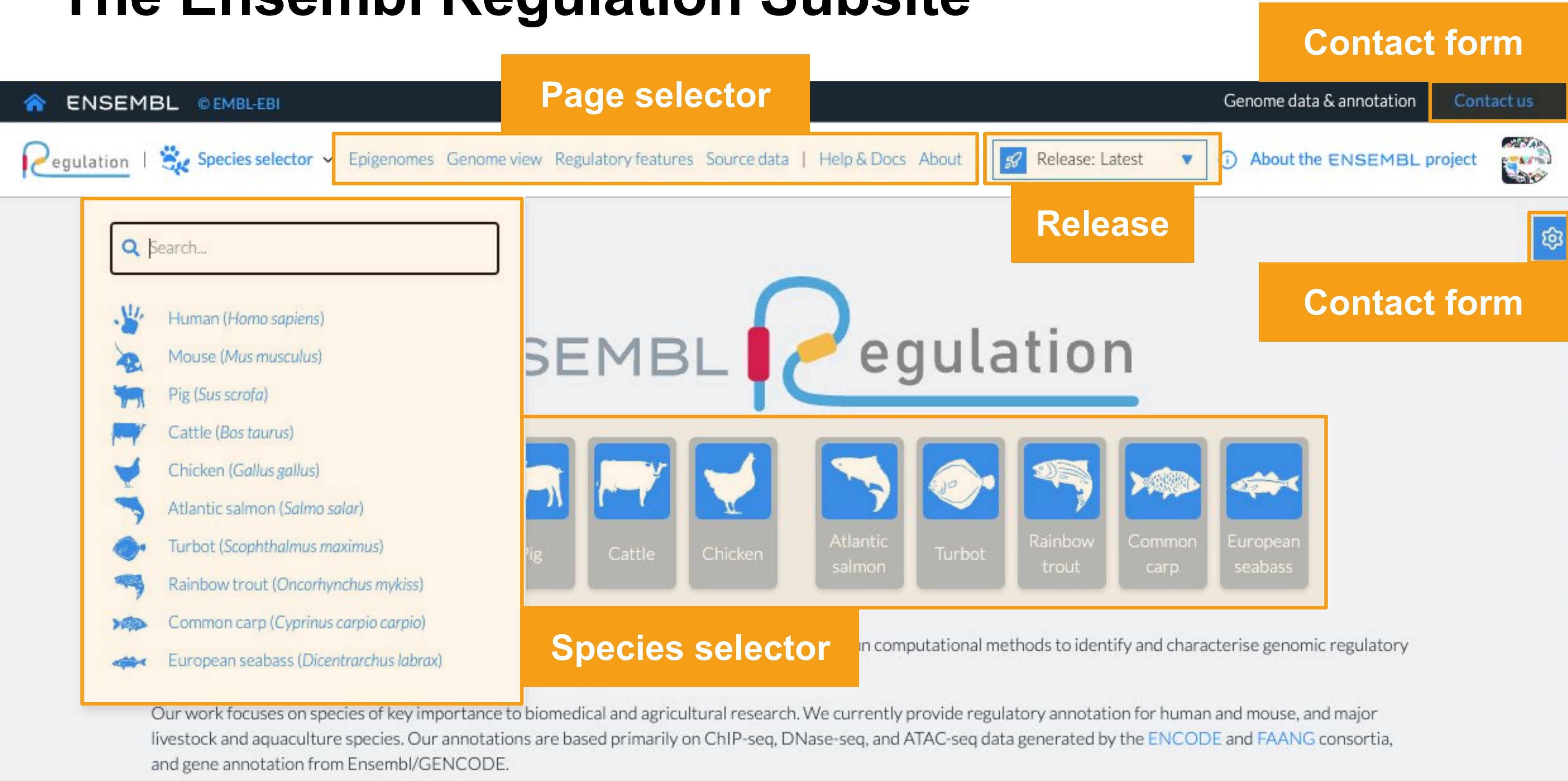
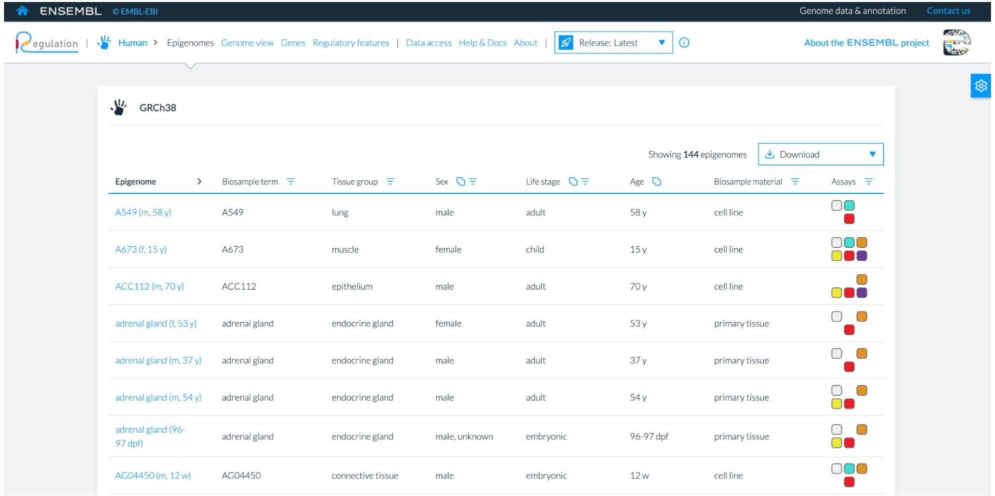
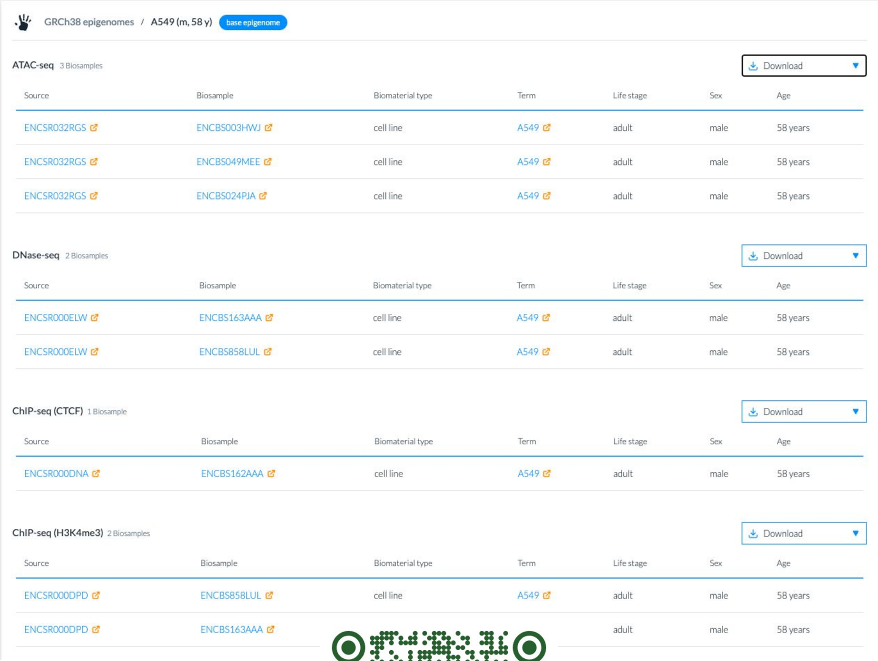
The Ensembl Regulation Subsite



The Ensembl Regulation Subsite: Epigenomes tab





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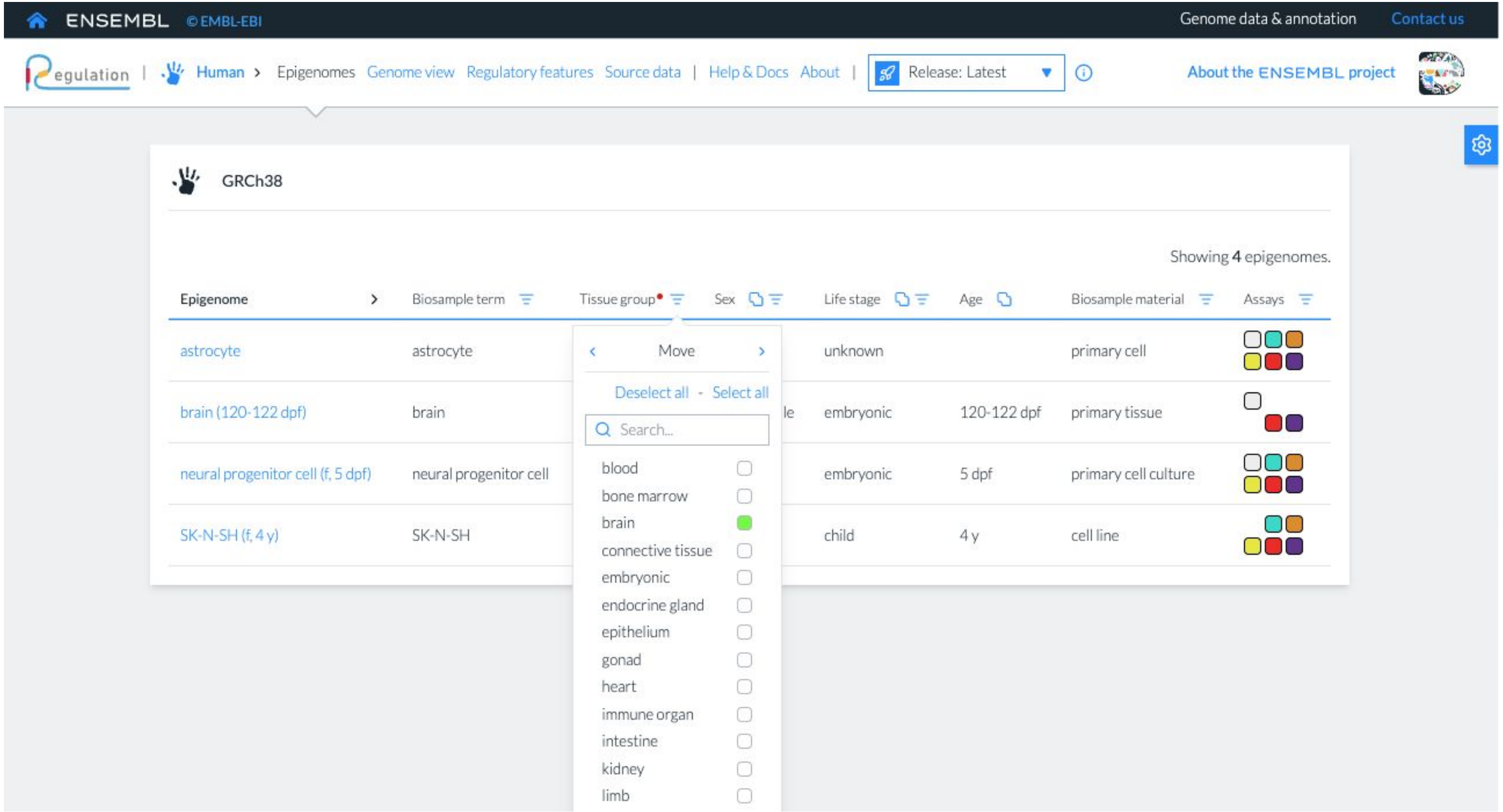
Epigenomes tab



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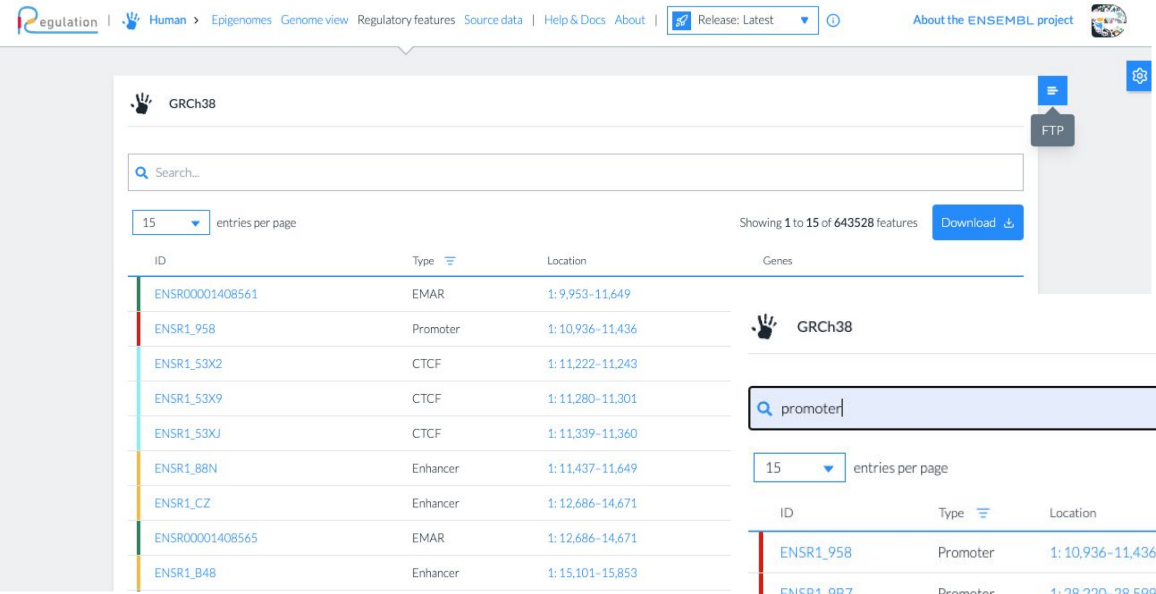
Regulatory Features tab



Regulation | Human › Epigenomes Genome view Regulatory features Source data | Help & Docs About | Release: Latest

About the ENSEMBL project

GRCh38

FTP

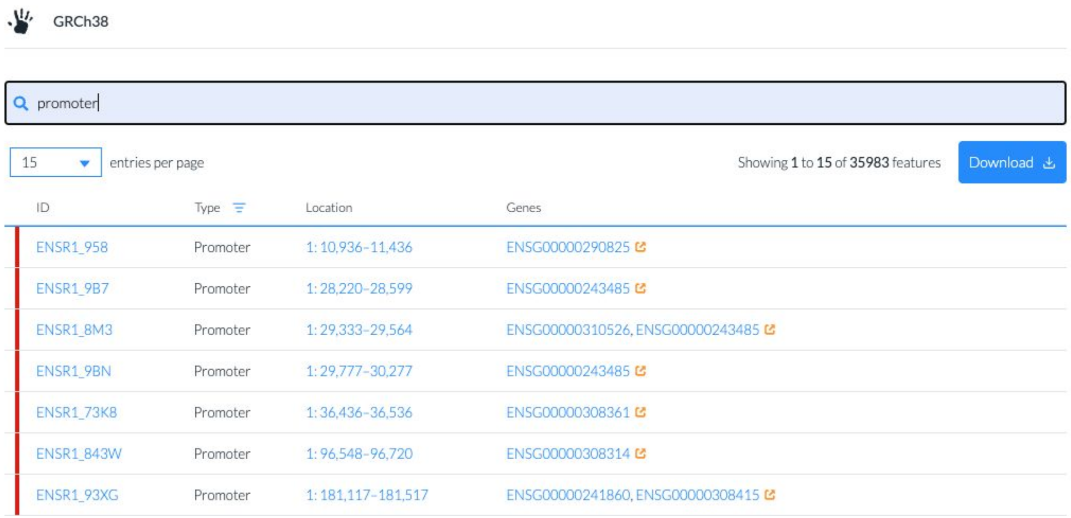
Search...

15 entries per page

Showing 1 to 15 of 643528 features

Download

ID	Type	Location	Genes
ENSR00001408561	EMAR	1: 9,953-11,649	
ENSR1_958	Promoter	1: 10,936-11,436	
ENSR1_53X2	CTCF	1: 11,222-11,243	
ENSR1_53X9	CTCF	1: 11,280-11,301	
ENSR1_53XJ	CTCF	1: 11,339-11,360	
ENSR1_88N	Enhancer	1: 11,437-11,649	
ENSR1_CZ	Enhancer	1: 12,686-14,671	
ENSR00001408565	EMAR	1: 12,686-14,671	
ENSR1_B48	Enhancer	1: 15,101-15,853	



GRCh38

promoter

15 entries per page

Showing 1 to 15 of 35983 features

Download

ID	Type	Location	Genes
ENSR1_958	Promoter	1: 10,936-11,436	ENSG00000290825
ENSR1_9B7	Promoter	1: 28,220-28,599	ENSG00000243485
ENSR1_8M3	Promoter	1: 29,333-29,564	ENSG00000310526, ENSG00000243485
ENSR1_9BN	Promoter	1: 29,777-30,277	ENSG00000243485
ENSR1_73K8	Promoter	1: 36,436-36,536	ENSG00000308361
ENSR1_843W	Promoter	1: 96,548-96,720	ENSG00000308314
ENSR1_93XG	Promoter	1: 181,117-181,517	ENSG00000241860, ENSG00000308415

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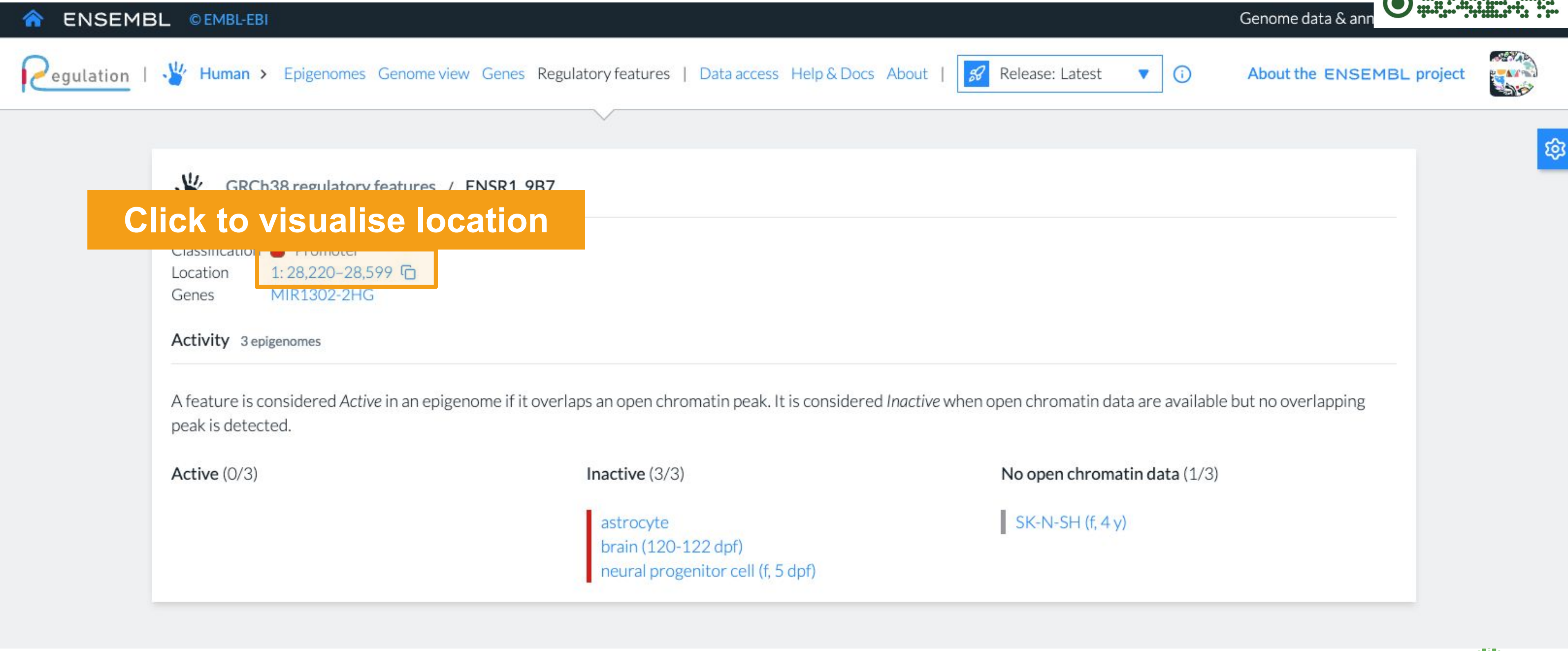
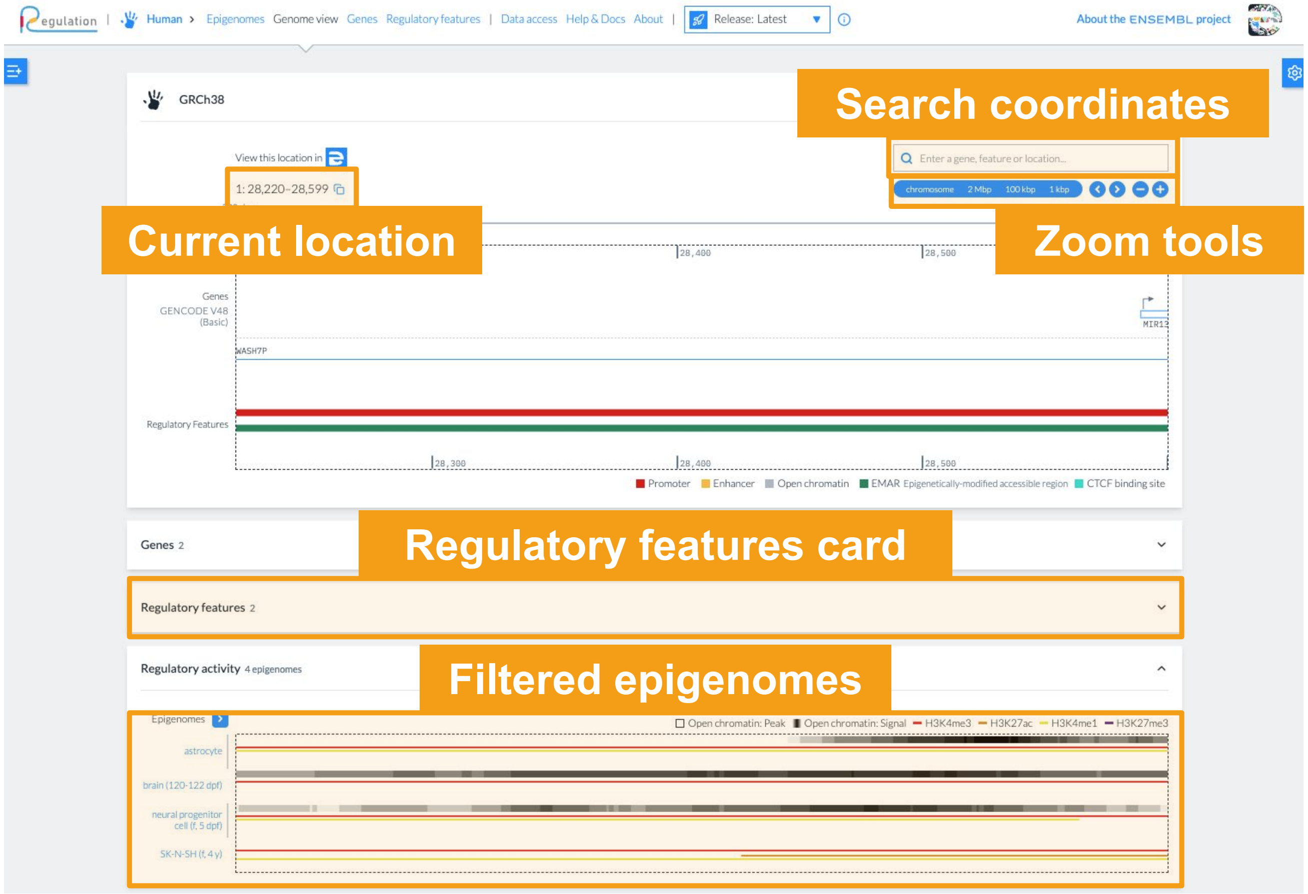
Regulatory Features tab



Genome view tab



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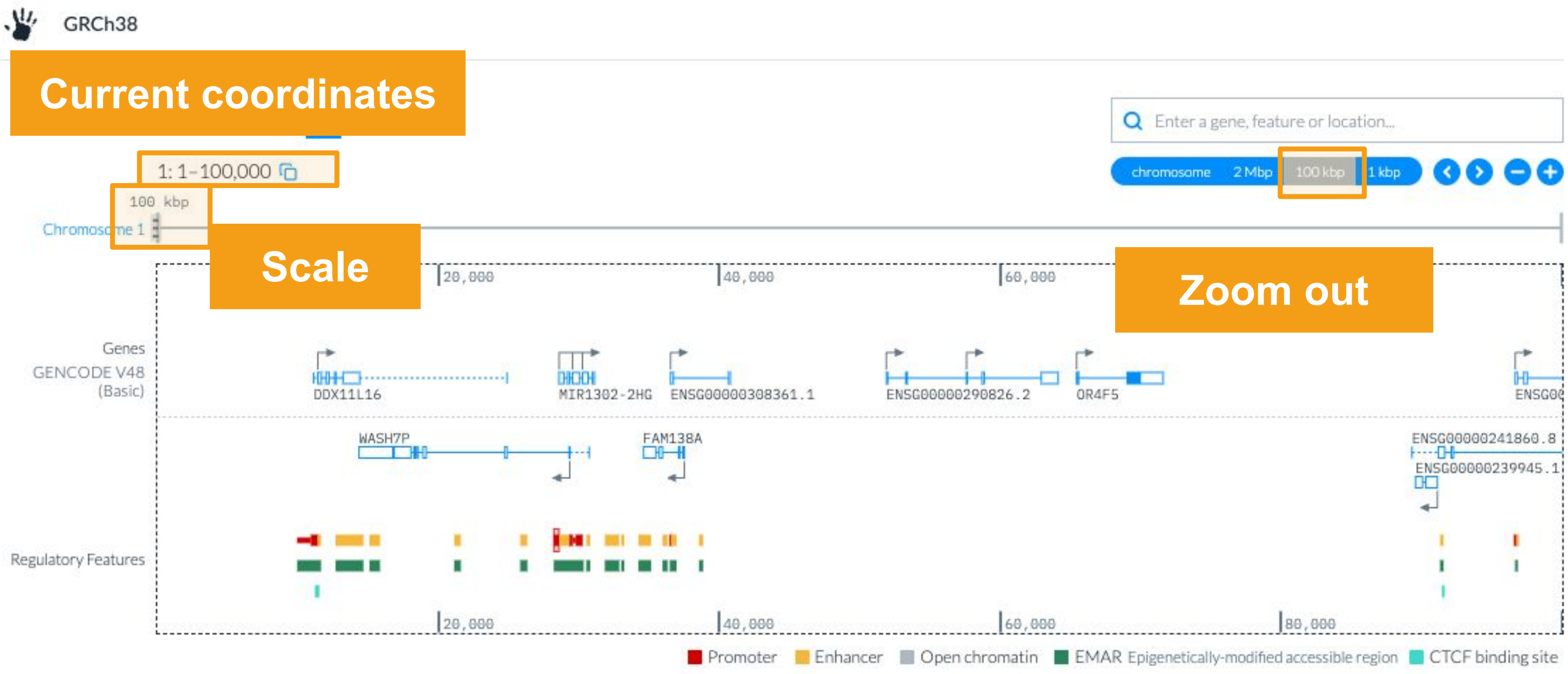
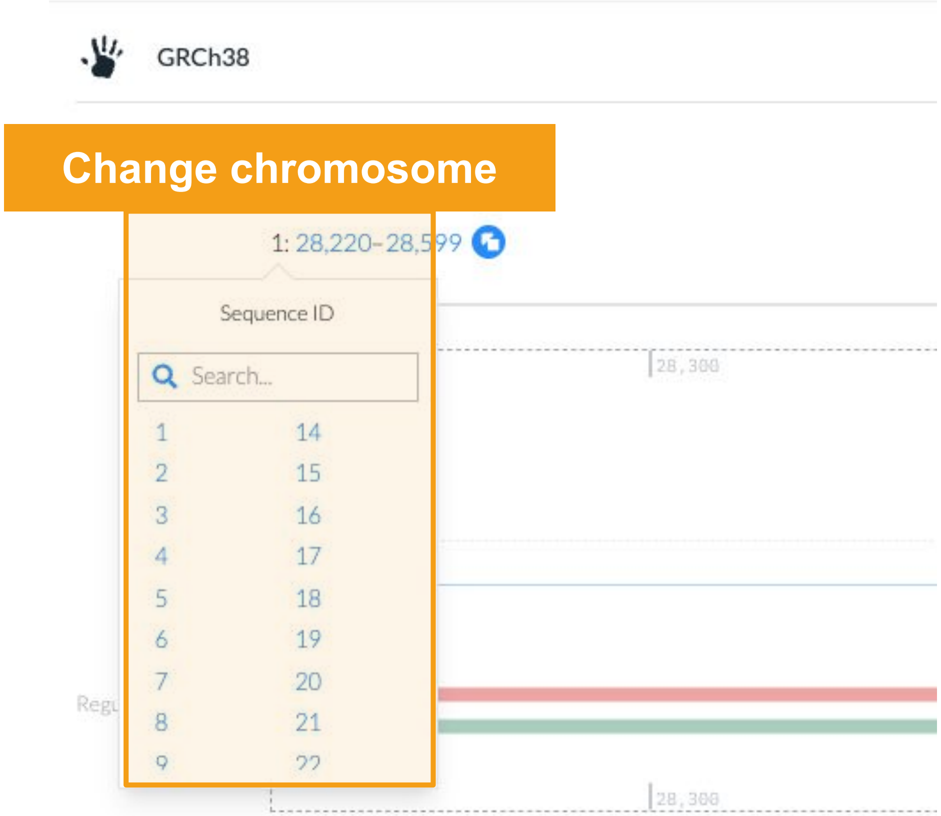
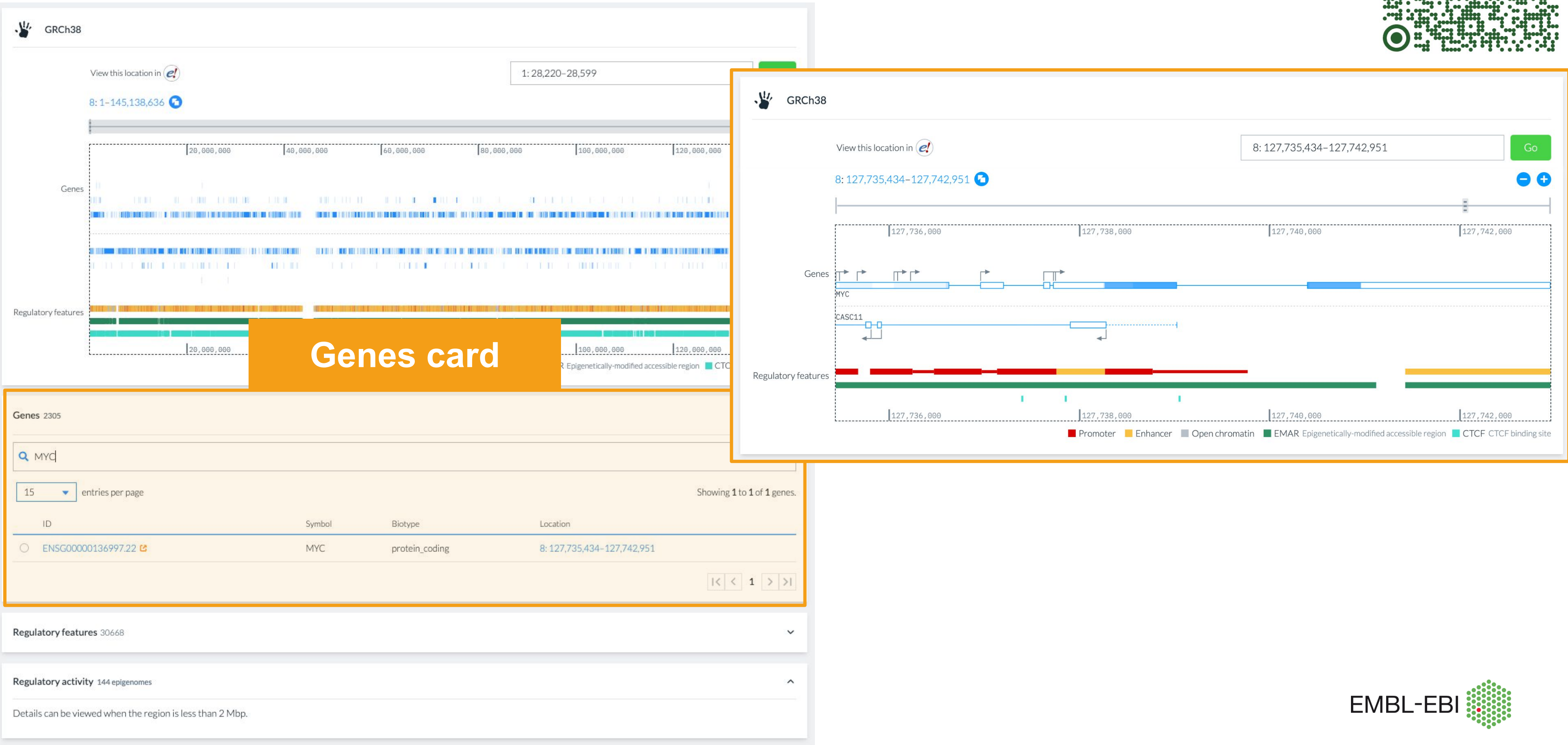
Genome view tab: zoomed out



Genome view tab: chromosome change



Genome view tab: Gene search



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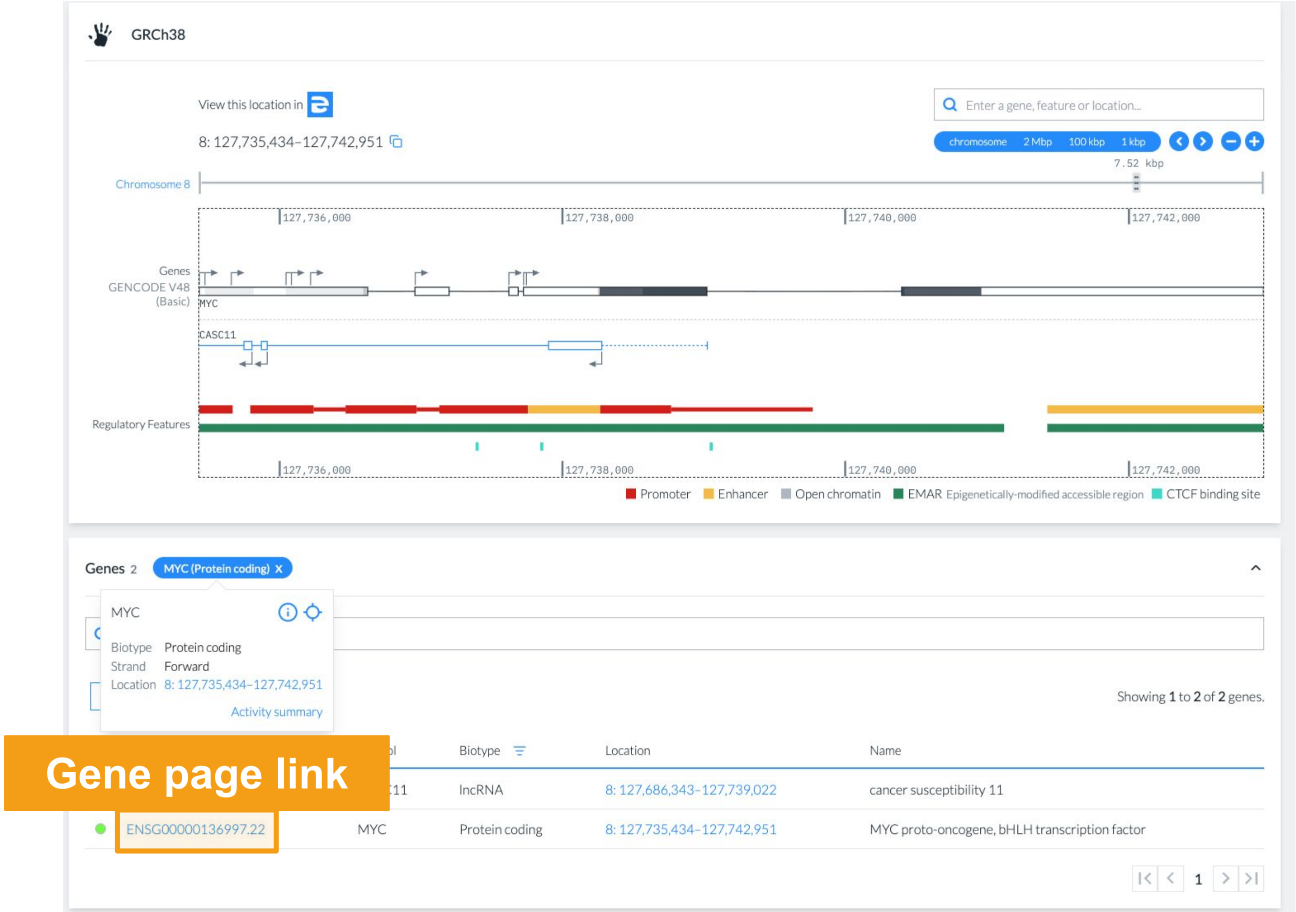
Genome view tab: Gene page link



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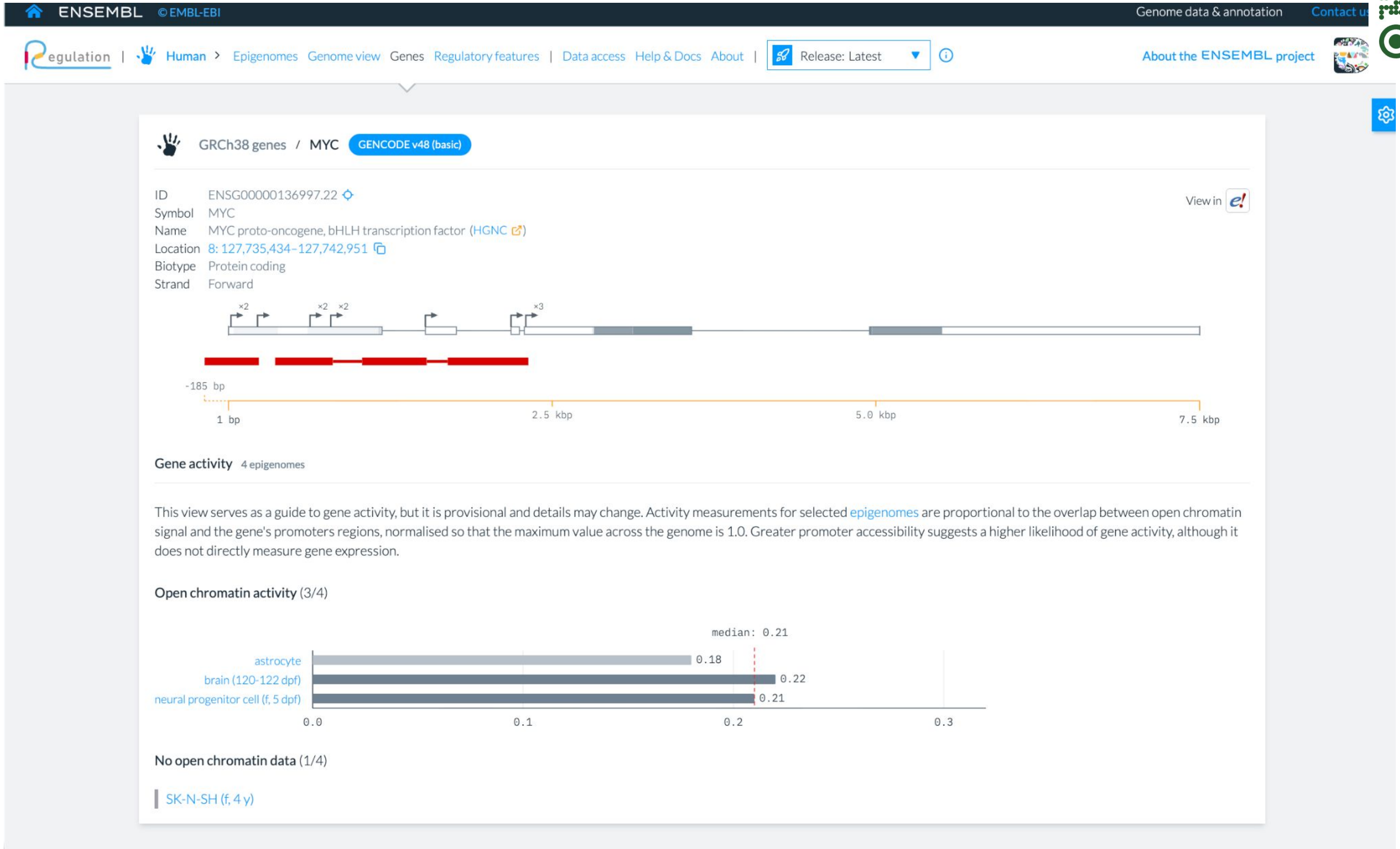
Genes tab



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GRCh38 genes / MYC
GENCODE v48 (basic)
ID ENSG00000136997.22
Symbol MYC
Name MYC proto-oncogene, bHLH transcription factor (HGNC)
Location 8: 127,735,434-127,742,951
Biotype Protein coding
Strand Forward
View in
×2
×2
×2
×3
-185 bp
1 bp
2.5 kbp
5.0 kbp
7.5 kbp
Gene activity 4 epigenomes
This view serves as a guide to gene activity, but it is provisional and details may change. Activity measurements for selected epigenomes are proportional to the overlap between open chromatin signal and the gene's promoters regions, normalised so that the maximum value across the genome is 1.0. Greater promoter accessibility suggests a higher likelihood of gene activity, although it does not directly measure gene expression.
Open chromatin activity (3/4)
median: 0.21
astrocyte
0.18
brain (120-122 dpf)
0.22
neural progenitor cell (f, 5 dpf)
0.21
0.0
0.1
0.2
0.3
No open chromatin data (1/4)
SK-N-SH (f, 4 y)

Genome view tab: Regulatory feature selection

Regulatory features selection

Genes 2 MYC (Protein co

Regulatory features 15 ENSR8_93TNWH ×

Search...

15 entries per page

Showing 1 to 6 of 6 features.

	ID	Type	Location	Genes
○	ENSR8_BXCG3	Promoter	8: 127,733,946-127,734,457	CASC11
●	ENSR8_93TNWH	Promoter	8: 127,735,249-127,735,669	CASC11, MYC
○	ENSR8_93TNWM	Promoter	8: 127,735,795-127,736,241	CASC11, MYC
○	ENSR8_93TNWT	Promoter	8: 127,736,468-127,736,968	MYC
○	ENSR8_BXCGJ	Promoter	8: 127,737,131-127,737,755	MYC
○	ENSR8_93TNXD	Promoter	8: 127,738,267-127,738,767	CASC11

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Genome view tab: Regulatory activity

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Genes tab

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GRCh38 GENCODE v48 (basic)

Search...

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Showing **1** to **15** of **54582** genes

ID	Symbol	Biotype	Location	Name
ENSG00000290825.2	DDX11L16	lncRNA	1: 11,121-24,894	DEAD/H-box helicase 11 like 16 (pseudogene)
ENSG00000310526.1	WASH7P	lncRNA	1: 14,356-30,744	WASP family homolog 7, pseudogene
ENSG00000243485.6	MIR1302-2HG	lncRNA	1: 28,589-31,109	MIR1302-2 host gene
ENSG00000237613.3	FAM138A	lncRNA	1: 34,553-37,595	family with sequence similarity 138 member A
ENSG00000308361.1		lncRNA	1: 36,526-40,778	
ENSG00000290826.2		lncRNA	1: 51,891-64,116	
ENSG00000186092.7	OR4F5	Protein coding	1: 65,419-71,585	olfactory receptor family 4 subfamily F member 5
ENSG00000241860.8		lncRNA	1: 89,295-181,719	
ENSG00000239945.1		lncRNA	1: 89,551-91,105	
ENSG00000308314.1		lncRNA	1: 96,638-105,742	
ENSG00000308579.1		lncRNA	1: 134,914-136,246	
ENSG00000239906.1		lncRNA	1: 139,790-140,339	
ENSG00000310528.1		lncRNA	1: 146,386-199,861	
ENSG00000241599.1		lncRNA	1: 160,446-161,525	
ENSG00000308415.1	DDX11L2	lncRNA	1: 180,931-184,937	DEAD/H-box helicase 11 like 2 (pseudogene)

1

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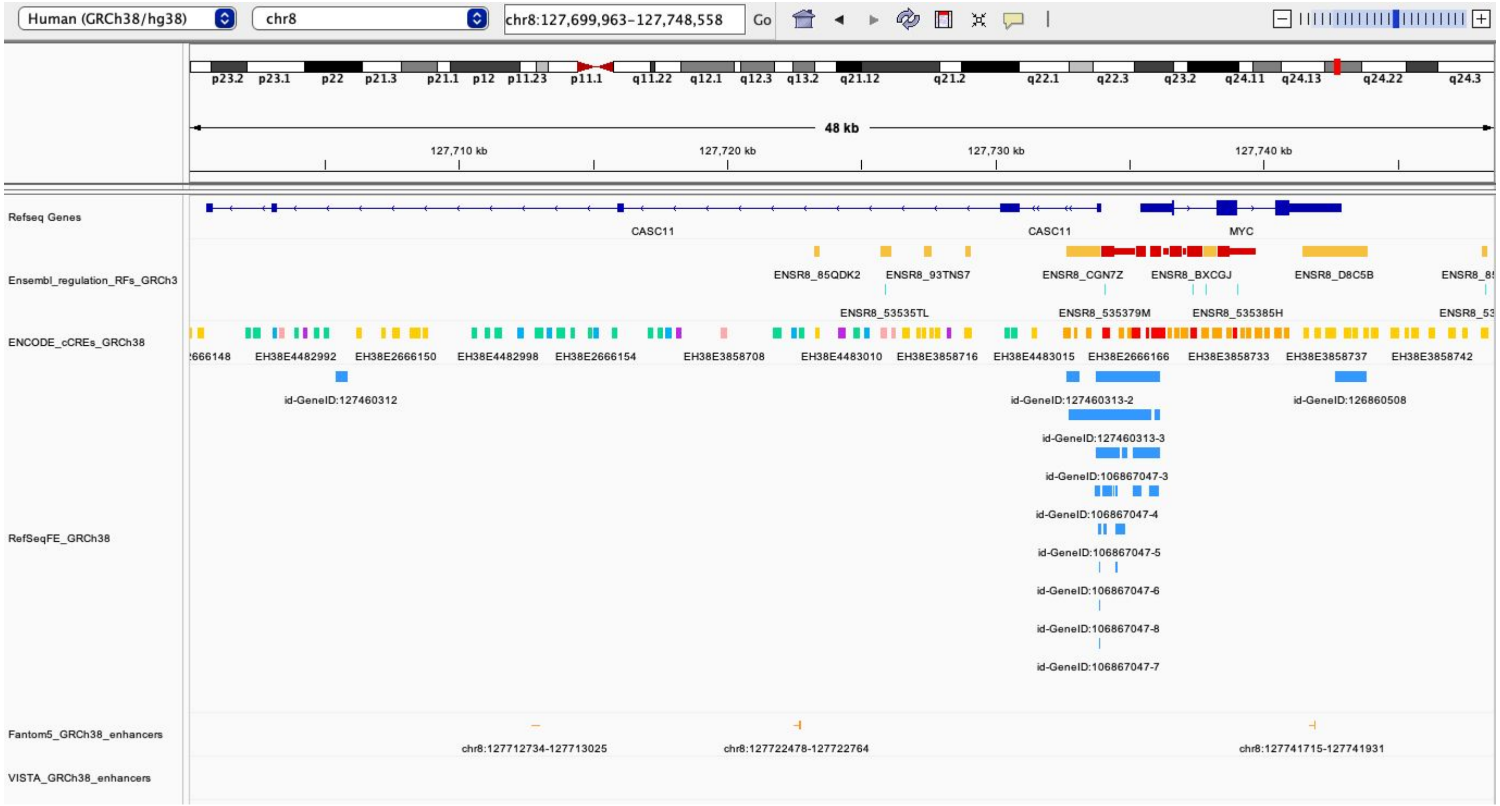
Differences between regulatory annotation frameworks

Resource	Main evidence categories	What it is useful for	Main caveat
Ensembl Regulation	Chromatin accessibility, histone modifications, CTCF ChIP-seq	Integrated regulatory features: promoters, enhancers, open chromatin regions, CTCF sites; activity across epigenomes	Evidence-based annotation, not direct proof of regulatory function
ENCODE cCREs / SCREEN	Chromatin accessibility, histone modifications, CTCF/TF ChIP-seq	Candidate cis-regulatory elements with promoter-like, enhancer-like, CTCF-like and other classes	Strong genome-wide catalogue, but still “candidate” elements
RefSeq Functional Elements	Manually curated experimental evidence, literature, protein binding, functional assays	High-confidence functional regions such as enhancers, silencers, insulators and locus control regions	Curated and evidence-rich, but less comprehensive genome-wide
FANTOM5	Transcription-based evidence, mainly CAGE	Active promoters and transcribed enhancers across many cell types/tissues	Best for active transcribed elements; may miss poised or non-transcribed CREs
Roadmap Epigenomics / ChromHMM	Chromatin accessibility, histone modifications	Chromatin-state annotations across many tissues/cell types	State labels are model-based and assay-dependent. Older technologies
VISTA Browser	Comparative genomics plus in vivo reporter assays	Experimentally validated developmental enhancers	Very valuable but focused on tested fragments, not comprehensive

Differences between regulatory annotation frameworks: GRCh38



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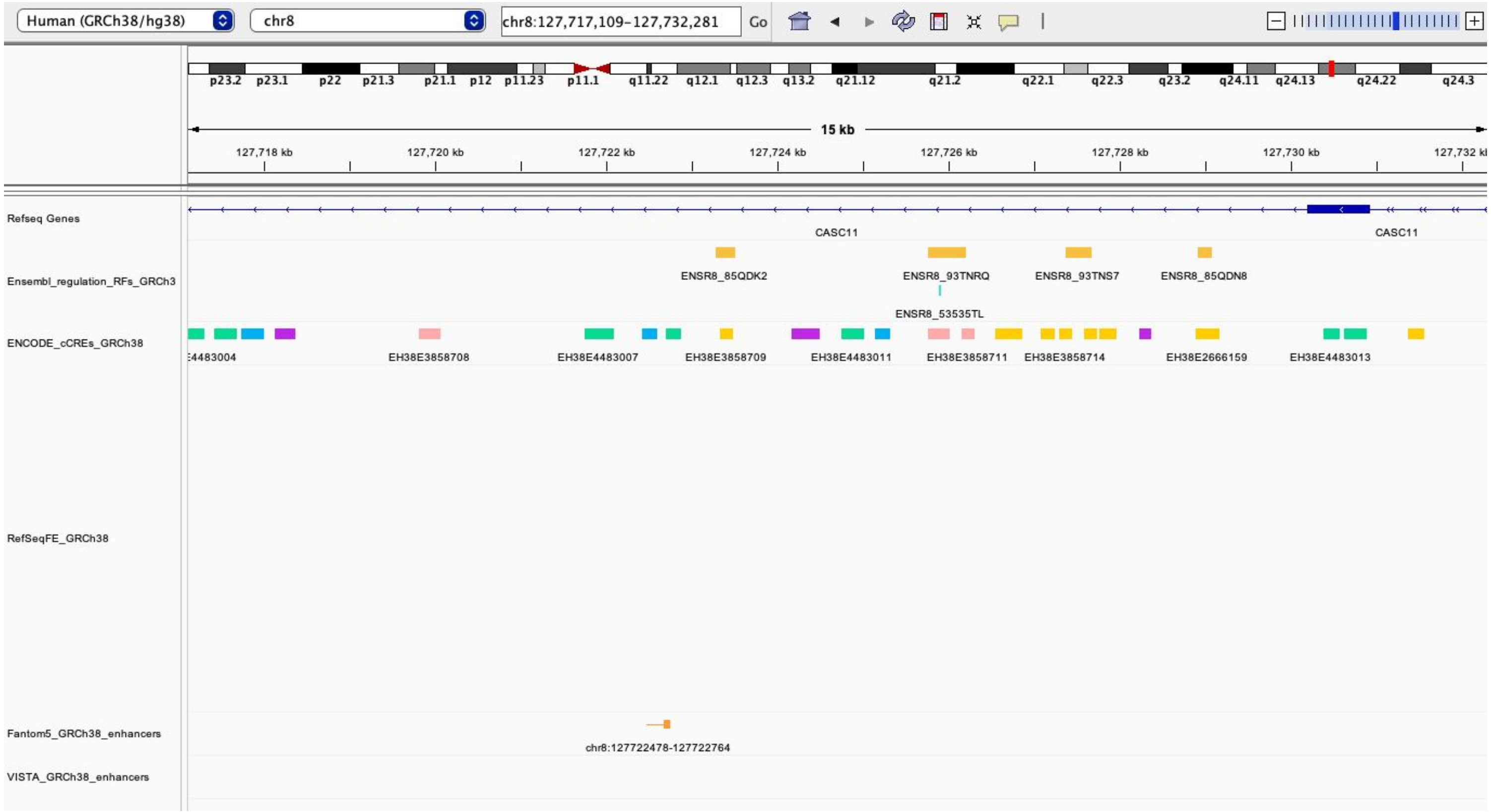
Differences between regulatory annotation frameworks: GRCh38



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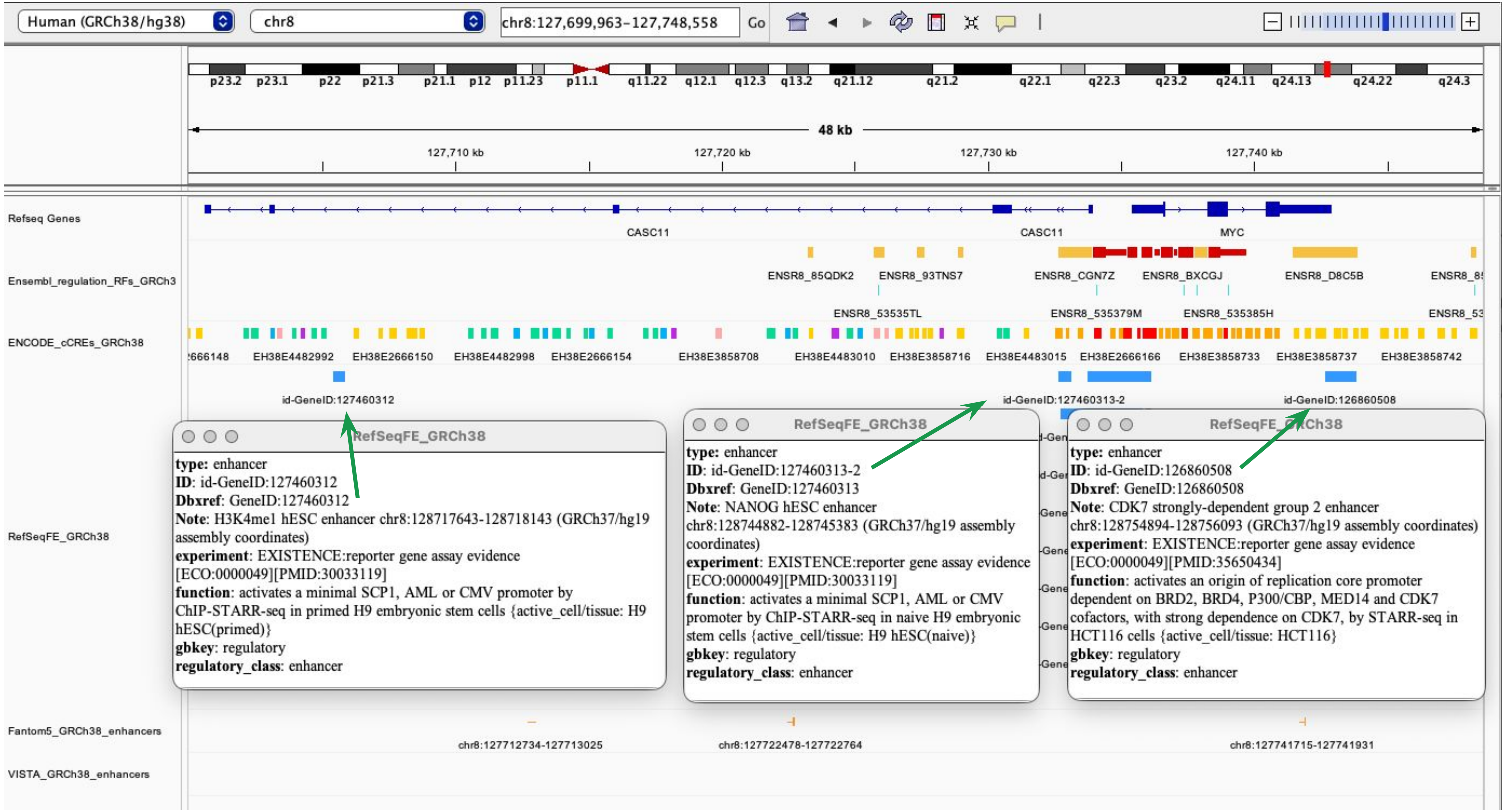
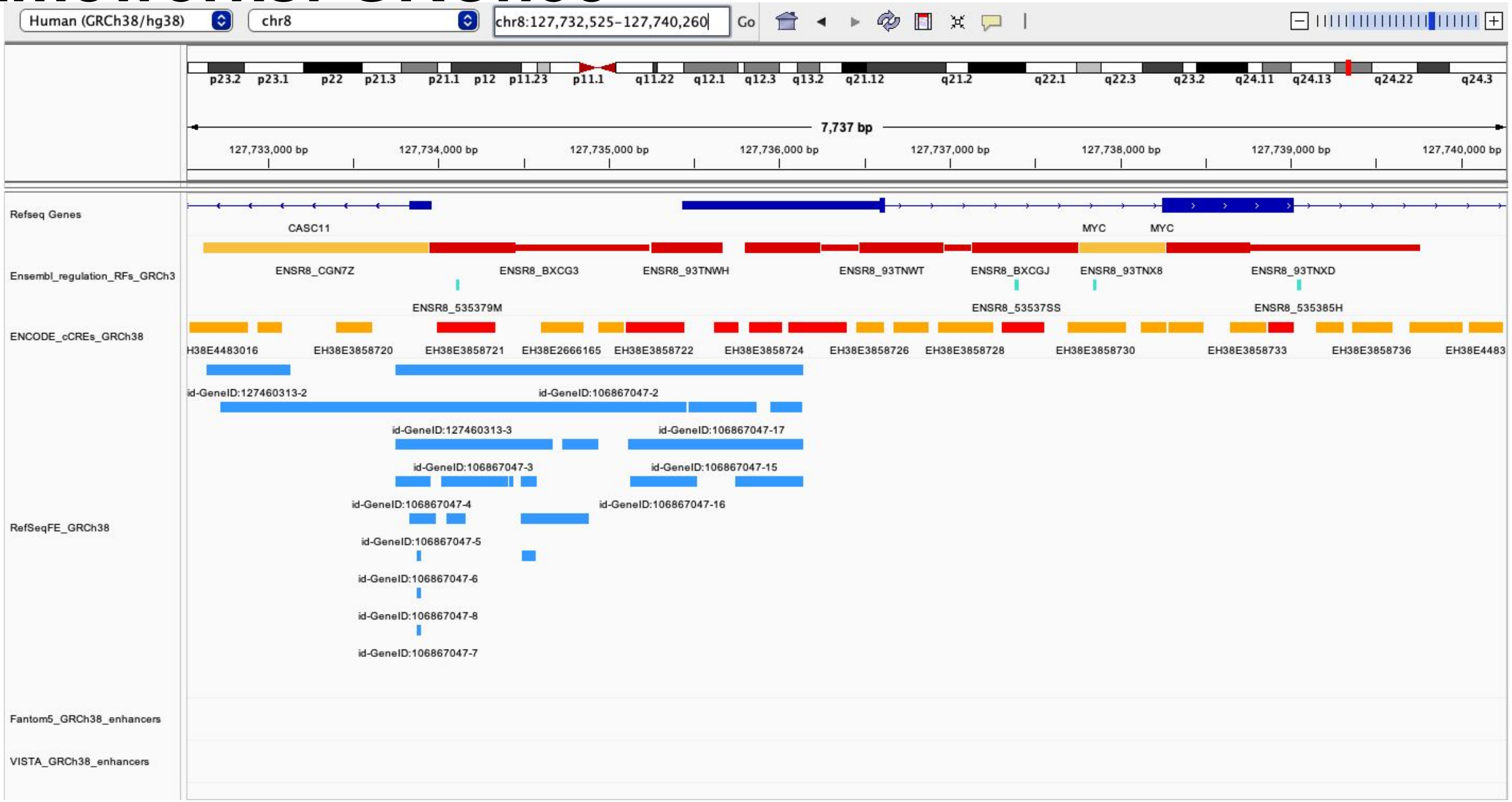
Differences between regulatory annotation frameworks: GRCh38



Differences between regulatory annotation frameworks: GRCh38



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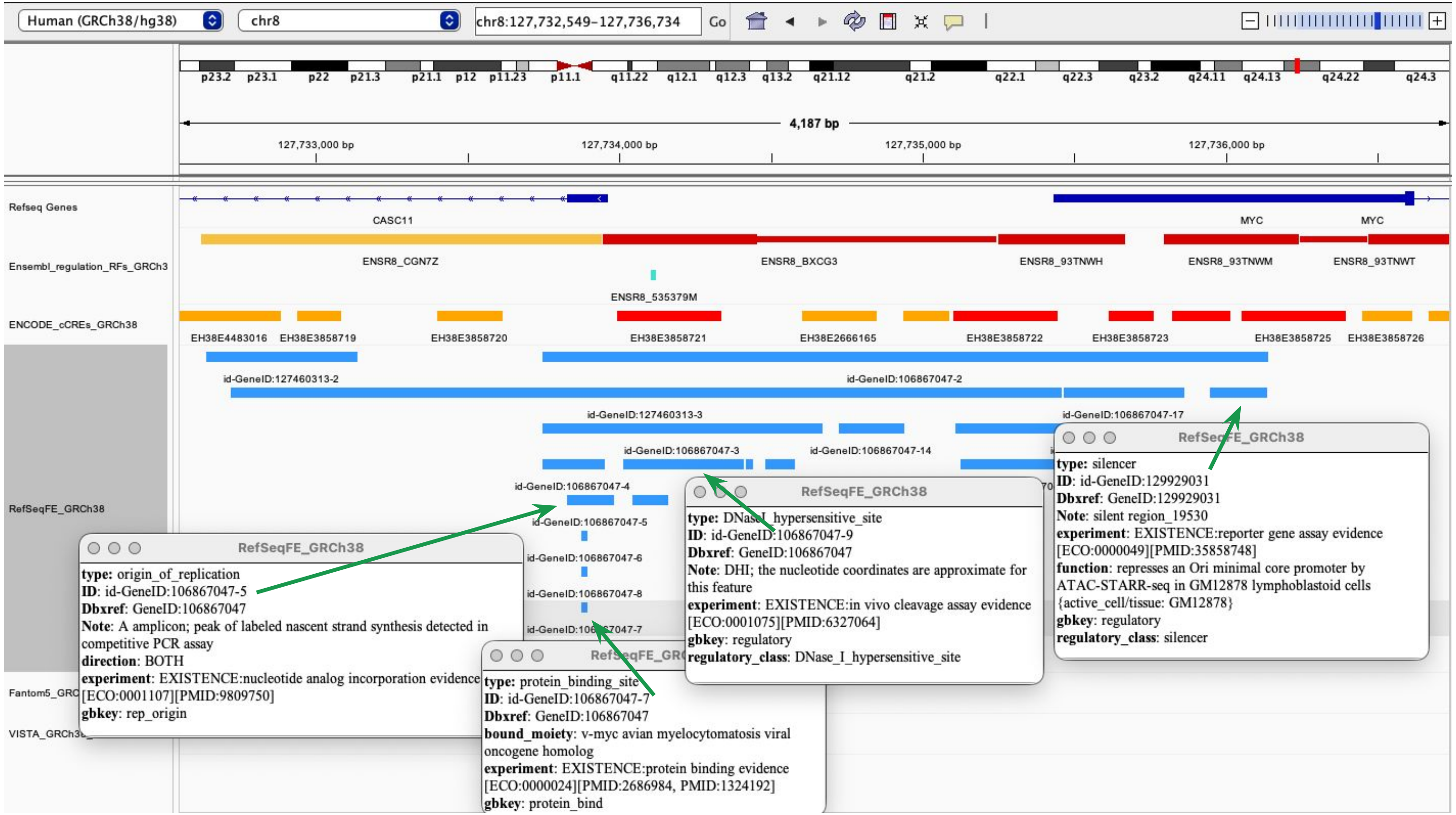
Differences between regulatory annotation frameworks: GRCh38



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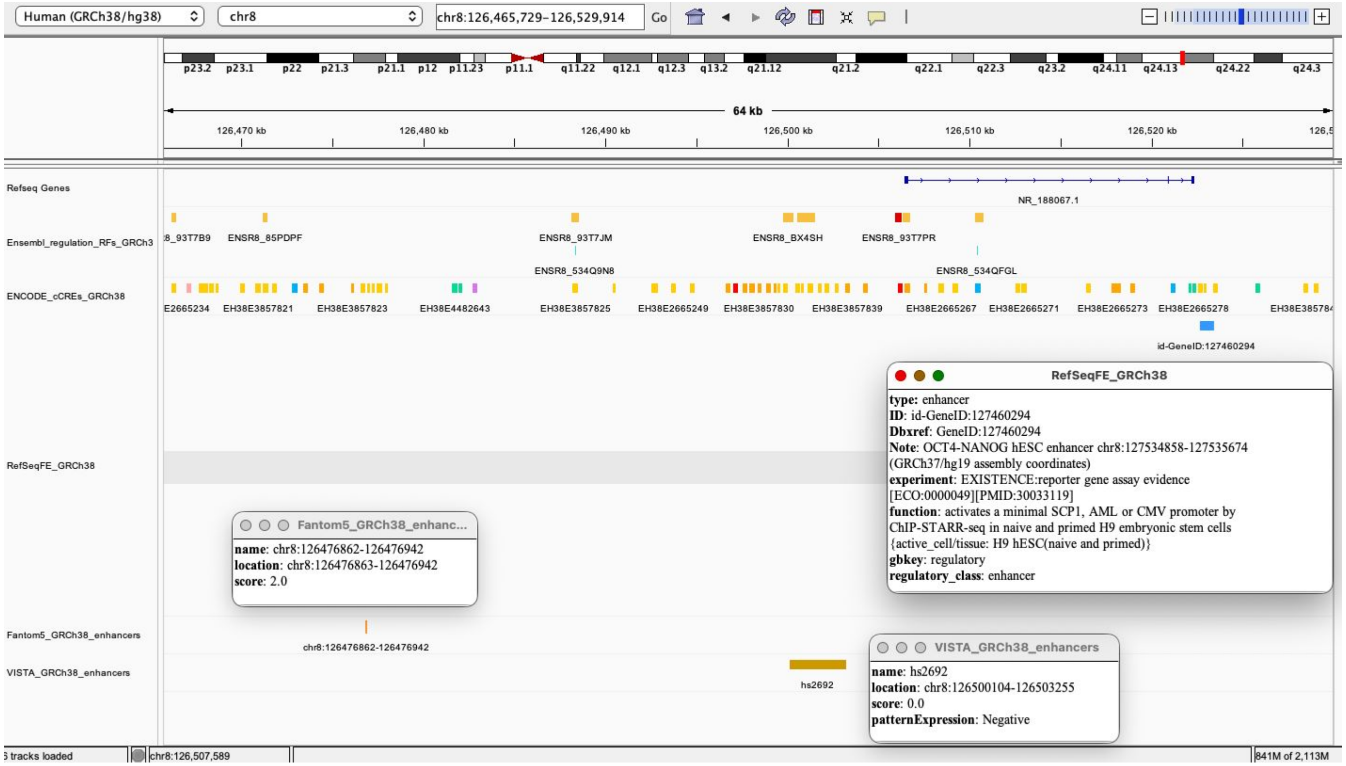
Differences between regulatory annotation frameworks: GRCh38



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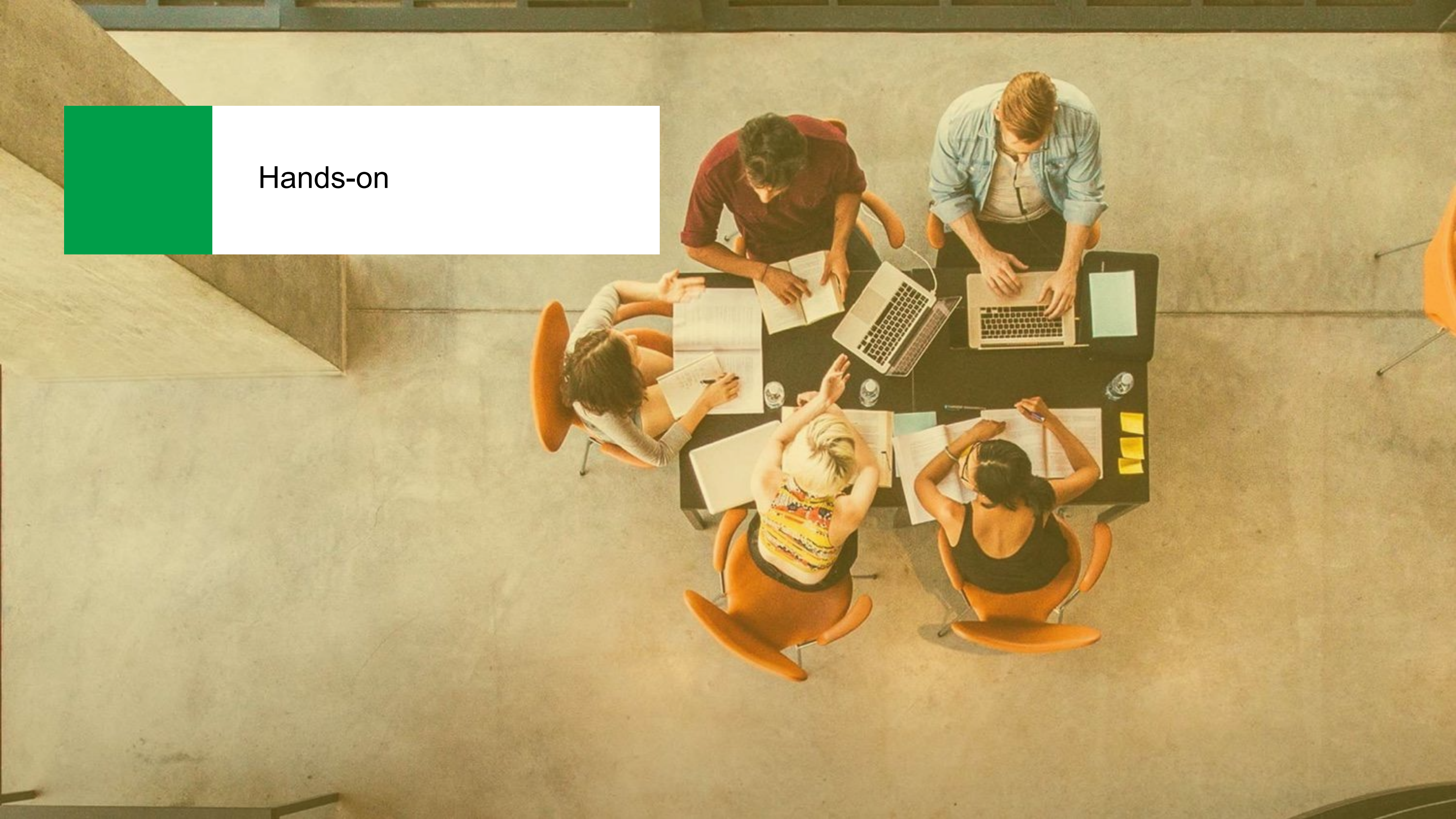
Questions?

Hands-on



Ensembl Regulation

Answer the following questions:

- ✓ Explore the **indicated locus** on the Genome View tab:
  - GRCh38, chr8:127,682,026-127,744,101
  - GRCm39/mm39, chr6:6,740,753-6,891,103
- ✓ Display and explore regulatory features using the Regulation card.
- ✓ Filter epigenomes based on the given criteria
- ✓ Compare regulatory activity across the group of selected epigenemes:
  - GRCh38: Caco-2 and HCT116 cell lines
  - GRCm39: Forebrain epigenomes
- ✓ Inspect supporting epigenomic evidence in the indicated epigenome
- ✓ Identify at least one context-dependent example.
- ✓ Compare the results with the ones obtained in the previous session.